

COVERAGE ASSURANCE FOR THE ESN

HOW CAN THE EMERGENCY SERVICES AND OTHER PUBLIC SAFETY ORGANISATIONS HAVE CONFIDENCE THAT AIRWAVE'S REPLACEMENT WILL PROVIDE THE NETWORK COVERAGE THEY EXPECT AND DESERVE?



BETWEEN 2018 AND 2020 THE UK'S EMERGENCY SERVICES AND OTHER PUBLIC SAFETY ORGANISATIONS WILL TRANSITION OFF THE TRUSTED AIRWAVE TETRA RADIO COMMUNICATIONS SYSTEM ONTO THE NEW EMERGENCY SERVICES NETWORK (ESN) RUNNING OVER EE'S ENHANCED COMMERCIAL LTE NETWORK. WITH SAFETY OF PUBLIC SAFETY USERS AND THE GENERAL PUBLIC PARAMOUNT TO A SUCCESSFUL TRANSITION, THERE SHOULD BE NO COMPROMISE IN NETWORK COVERAGE OR SERVICE QUALITY.

PUBLIC SAFETY LTE & THE ESN

The Home Office's Emergency Service Mobile Communications Programme (ESMCP) is responsible for providing the next generation communications system for the police, fire, ambulance and other public safety organisation. This system is called the ESN and will provide next generation mission critical voice and broadband data services for these users.

Unlike today's Airwave system which uses a dedicated radio network specifically designed for public safety users, the ESN will use a mobile-data (LTE) based technology using the shared infrastructure of the EE commercial network which will be enhanced for mission critical requirements

including full national coverage. Motorola will provide end to end systems integration and public safety functionality.

ESN users will benefit from mission critical grade data services such as real time video streaming, multi-media video dispatch, and location-based services. All of which

supplement and enhance the mission critical voice service thus providing improved situation awareness for both first responders and dispatchers.

ESMCP's intent is to transition users from Airwave to the ESN using a staggered approach based on geographic region¹. The Home Office

plans indicate the first emergency services organisations will have transitioned onto the ESN by mid-2018 and all organisations across Great Britain to have completed transition before the end of 2020. Based on the lessons learnt from the

"The promise of ESN is that it will be at least as good as Airwave for voice and better for data" [Home Office]

original transition onto the Airwave network, this is an extremely challenging transition timetable which when combined with ESMCP trail blazing new technologies, makes the ESN an extremely ambitious programme.

This brochure provides an overview of the network coverage and service assurance challenges emergency service organisations should consider in advance of transitioning from Airwave onto the ESN.

ESN COVERAGE AND SERVICE ASSURANCE

With up to 300,000 public safety users and 50,000 vehicles planned for the new service, there cannot be any compromise to the quality or reliability of the ESN – mission critical comms are just that, mission critical, and any failures in the

comms service could result in serious injury or death for public safety personnel or members of the public.

Chief Constables and employers of other public safety staff have a legal *duty of care*² to take reasonable steps to ensure the safety of their staff, with there being severe

consequences if they neglect to do so. Sign off for accepting the ESN service into live operation ultimately rests with the end-user organisation. It is therefore vital that they take all steps to satisfy themselves that the ESN will be *at least as good as Airwave* in terms of safety and functionality.



1. National Audit Office report into "Upgrading emergency service communications: the Emergency Service Network" – Figure 12 (September 2016)

2. Health and Safety at Work Act 1974. See section 51A specifically in relation to Chief Constables

3. NAO report section 3.8

So what does this mean in practice?

ESMCP will take responsibility for overseeing the comprehensive testing of ESN functionality, whereby EE and Motorola conduct initial testing themselves and ESMCP (supported by KBR) provide assurance of the design and implementation. The services will then be tested in a live environment through field trials and regional pilots undertaken by emergency services, supported by suppliers and assured by ESMCP.

Unlike the comprehensive drive testing and validation of coverage conducted on every road before the Airwave service went live, ESMCP's plans³ to test coverage appear limited to *"undertake limited drive-testing, just sufficient to assure the programme that EE's prediction models are accurate"*. This approach does not however address the fundamental end-user requirement of comprehensive nationwide testing of network coverage which is

unique to every region and which will require assurance through testing at a localised level whether it be on roads, railways or areas only accessible on foot. The expectation being that the end-user organisations will conduct these tests before going live on the ESN.

"We are clear that we won't take any risks with public safety and there will be no gap in the emergency services' communications provision"

[Minister for Policing and Fire Service, April 2017]

EXPERIENCE AND EXPERTISE MATTER

When it comes to accepting the service for live operation and pulling the plug on Airwave, end-users need to have complete confidence that the ESN will deliver... everywhere. That is why having comprehensive drive testing and validation of network coverage is paramount.

AWTG has been coverage testing and optimising mobile networks since 2006. As a trusted partner for regulators and mobile operators around the world, AWTG has proven

credentials in conducting independent benchmarking tests (including voice-over-LTE) tailored around the specific requirements of its clients.

AWTG's experience is well suited to testing ESN environments, whether it be through conducting drive testing in urban and rural environments, or to testing network coverage in shopping centres, pedestrianized zones, stations, tunnels, ports, airports, railway lines and within buildings. AWTG also has a thorough understanding of ESMCP, the ESN and the needs of the end-user organisations, having worked with the National Audit Office to produce the report on the ESN.

For areas of poor indoor coverage AWTG has an in-house team of engineers who can cost effectively design and install inbuilding indoor coverage solutions if required.

11 years of optimising mobile operators' network coverage means that AWTG is well placed to work with EE, ESMCP and emergency service organisations to ensure any gaps uncovered in ESN coverage can be effectively resolved.





About AWTG:

AWTG is a leading telecommunications services and IoT platform provider specialising in Networks, Smart Infrastructure, IoT and M2M solutions. We deliver a full range of business to business and network related services that are both cost effective, scalable and deliver rapid return on investment. AWTG works with our clients to quickly identify needs, design and deploy efficient solutions, drive benefit realisation and provide on-going maintenance and customer care services.

Founded in 2006, AWTG provides the highest degree of proficiency, flexibility and on demand scalability to help clients lower costs, derive value by streamlining operations and ultimately improve productivity.

For more information on AWTG's coverage assurance testing, consulting and in-building coverage services, please contact: 0203 515 5151 or enquiries@awtg.co.uk

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